

LAB | Apache Airflow Foundations Lab

Lab for today : Q1 , Q2 and only the MCQ

Q. 1 — Explain Workflow orchestration

Explain the concept in plain language. Include:

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- its main purpose;
- where it fits in the lesson workflow;
- one mistake a new learner should avoid.

Workflow orchestration means automatically managing a series of tasks so they run in the correct order.

Main purpose: To automate workflows and to make sure each task runs at the right time and only after the previous task is finished.

Where it fits in the lesson: It is the main idea behind Apache Airflow. Airflow uses a DAG to define the workflow and manages the tasks from start to finish.

Common mistake: Thinking Airflow does the data processing.

Q2 — Connect Workflow Orchestration and Apache Airflow

- Explain the role of each concept.
- State how they are related.
- Identify one reason they should not be treated as interchangeable.

Workflow orchestration decides what should run, when it should run, and what should run next. Without orchestration, people would need to run scripts manually or use separate timers.

Apache Airflow is a platform for defining, scheduling, monitoring, and retrying workflows. Workflows are written as DAGs (Directed Acyclic Graphs).

How they are related: Workflow orchestration is the concept, and Apache Airflow is a tool that implements that concept.

Why they are not interchangeable: Workflow orchestration is a general process, while Apache Airflow is one specific software platform used to perform it.

Q. 5 — MCQ: Purpose of Workflow orchestration: A

Q. 6 — MCQ: Applying Apache Airflow: B

Q7 — Explain Scheduler

Scheduler is the Airflow component that decides when tasks should run. It checks the DAG and schedules tasks that are ready to execute.

Main purpose: To monitor DAGs and schedule tasks at the correct time.

Where it fits in the lesson workflow: The Scheduler reads the DAG, checks task dependencies, and sends ready tasks to the Executor.

Q. 11 — MCQ: Purpose of Scheduler: A

Q. 12 — MCQ: Applying Webserver and Airflow UI: B

Q. 17 — MCQ: Purpose of Executor A

Q. 18 — MCQ: Applying DAG B

Q. 23 — MCQ: Purpose of Operator: A

Q. 24 — MCQ: Applying DAG file structure B

Final Recap

Before finishing, confirm that you can:

- Explain the main purpose of Workflow orchestration and Main Airflow components.
- Connect Scheduler and Metadata database to the wider workflow.
- Apply Executor and Task and task instance to a bounded BritMart scenario.
- Validate a result involving Operator and Task dependencies.
- Complete a practical or design task without guessing missing requirements.
- Explain one trade-off rather than presenting a tool as universally correct.
- Retain evidence that another learner can review.
- Correct a mistake after comparing your work with the suggested answer.